

Gray Reid

COMPUTATIONAL MODELING AND MACHINE LEARNING, NYC

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Profile

- PhD graduate in computational physics with expertise in machine learning and scientific computing, specializing in algorithm development for large-scale simulations. Proven track record in enhancing computational efficiency and accuracy over state-of-the-art techniques. Eager to transition into a Industry ML role to apply advanced modeling skills and drive innovation in tech. Committed to leveraging strong research background to contribute to high-impact projects.

Skills

Machine Learning

- Neural Networks, Deep Learning, Reinforcement Learning
- Attention Mechanisms, Transformers, CNNs, RNNs, GANs, Diffusion Models, GRUs, LSTMs

Scientific Computing

- HPC, Numerical Methods, Computational Modeling
- Finite Difference, Finite Element, Finite Volume, Spectral Methods, Monte Carlo, AMR, Multigrid
- Optimization, Profiling, Debugging

Mathematics and Physics

- Calculus, PDEs, SDEs, ODEs, Linear Algebra, Probability, Statistics
- Lagrangian Mechanics, Hamiltonian Mechanics
- Quantum Mechanics, Statistical Mechanics, Thermodynamics
- Optics, Spectroscopy, Computational Optics
- Field Theory, Fluid Dynamics, Electromagnetism, General Relativity

Languages and Frameworks

- C, Python, Fortran
- Pytorch, Jax, TensorFlow
- MPI, OpenMP, CUDA

Numerical Analysis

- Error analysis, Stability Analysis, Complexity Analysis, Convergence Analysis
- Conditioning, Discretization

Research

- Research Design, Scientific Writing, Literature Review
- Data Analysis, Data Visualization
- Mentoring, Curriculum Development, Presentations

Select Awards

NSERC CGSD

Canada's premier doctoral fellowship



UBC

Four Year Fellowship

Doctoral funding based on academic excellence



UBC

NSERC CGSM

Canada's premier masters fellowship



UBC

Governor General's Medal

Canada's highest academic honor



Acadia

Education

PhD Physics

Vancouver, BC, Canada

UNIVERSITY OF BRITISH COLUMBIA

2024

- Improved resolution and accuracy of high performance simulations of gravitational collapse by orders of magnitude over state-of-the-art without significantly increasing computational cost
- Developed expertise in computational modeling, numerical methods, high performance computing, debugging and optimization of massively parallel systems

BSc Honours Physics

Wolfville, NS, Canada

ACADIA UNIVERSITY

2012

- Highlighted deficiencies in existing techniques for modeling TEM imaging of materials and developed new methods for addressing these deficiencies
- Developed improved equations describing cavity ring-down spectroscopy for use in chemical sensing applications

Select Publications

Universality in the Critical Collapse of the Einstein-Maxwell System



PHYSICAL REVIEW D

2023

- Improved resolution and accuracy by several orders of magnitude over state of the art
- Improved resolution and accuracy of in-lab adaptive mesh refinement (AMR) software for simulation of generic PDEs
- Resolved conflicts with previous investigations and provided a framework for understanding discrepancies in related literature.

Reference Metric Approach to the Z4 System



PHYSICAL REVIEW D

2023

- Achieved performance equivalent or exceeding leading formulations of General Relativity such as GBSSN and FCCZ4
- Derived a novel hyperbolic formulation of General Relativity using a reference metric approach
- Demonstrated the stability of the formulation through a pseudodifferential first order reduction and simulations in the strong field regime

Stability of Nonminimally Coupled Topological-Defect Boson Stars



PHYSICAL REVIEW D

2024

- Performed both fully non-linear and semi-analytic stability analysis for stationary solutions in f(R) gravity
- Resolved the open question of stability for topological-defect boson stars in f(R) gravity